MILESTONE 4 PRESENTATION

Multimodal AI Assistant for Smart Agriculture

Crop Disease Detection + RAG Advisory System for Indian Farmers

Crops
Corn · Potato · Rice
Wheat · Sugarcane

Classes
17 disease categories

Images
13,324 total

Model
MobileNet V3 Large

DATASET OVERVIEW

Kaggle — kamal01

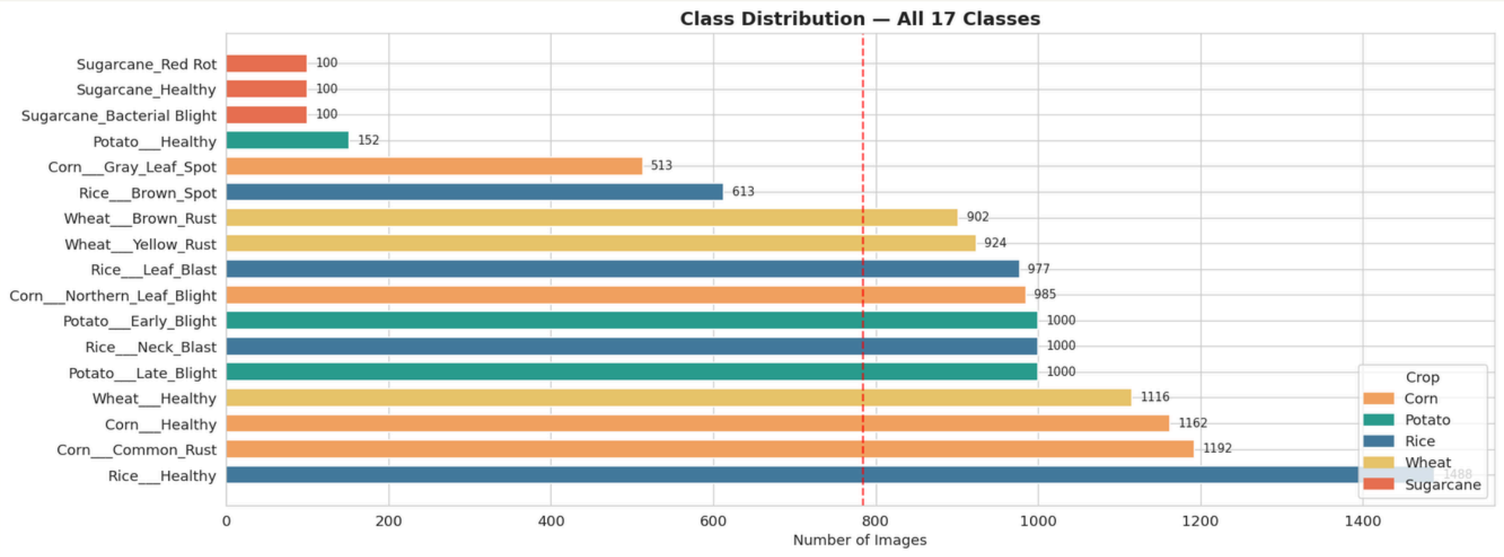
13,324
Total Images

17
Disease Classes

5
Crop Species

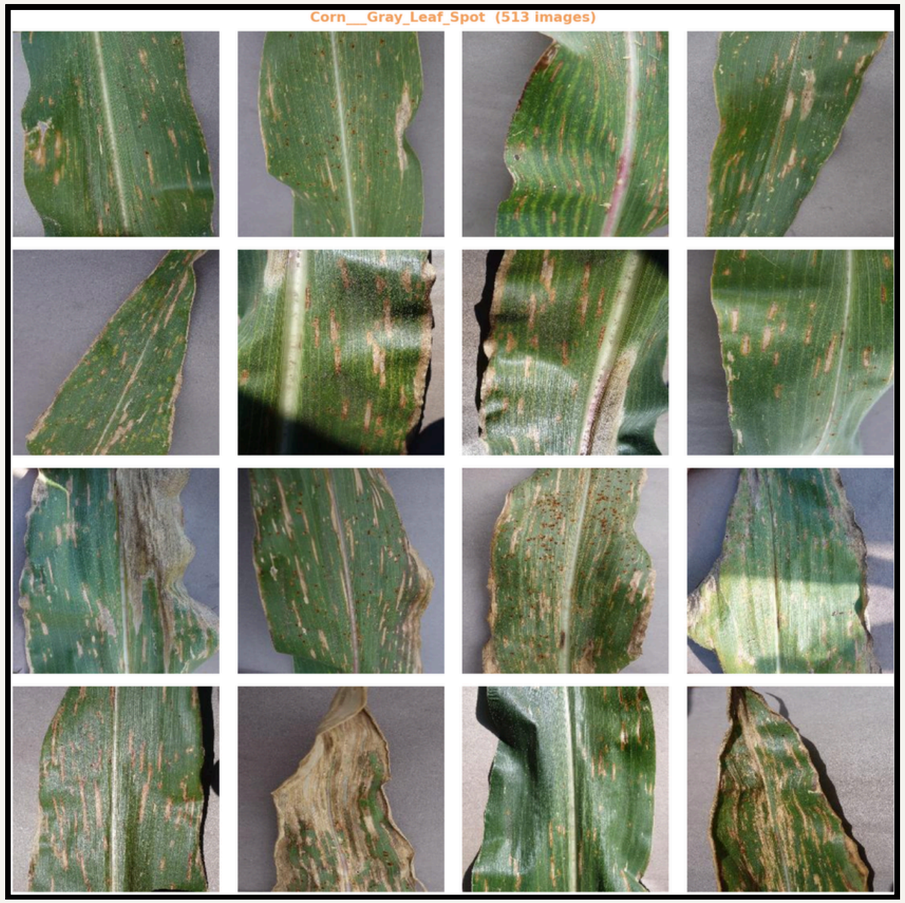
14.9×
Imbalance Ratio

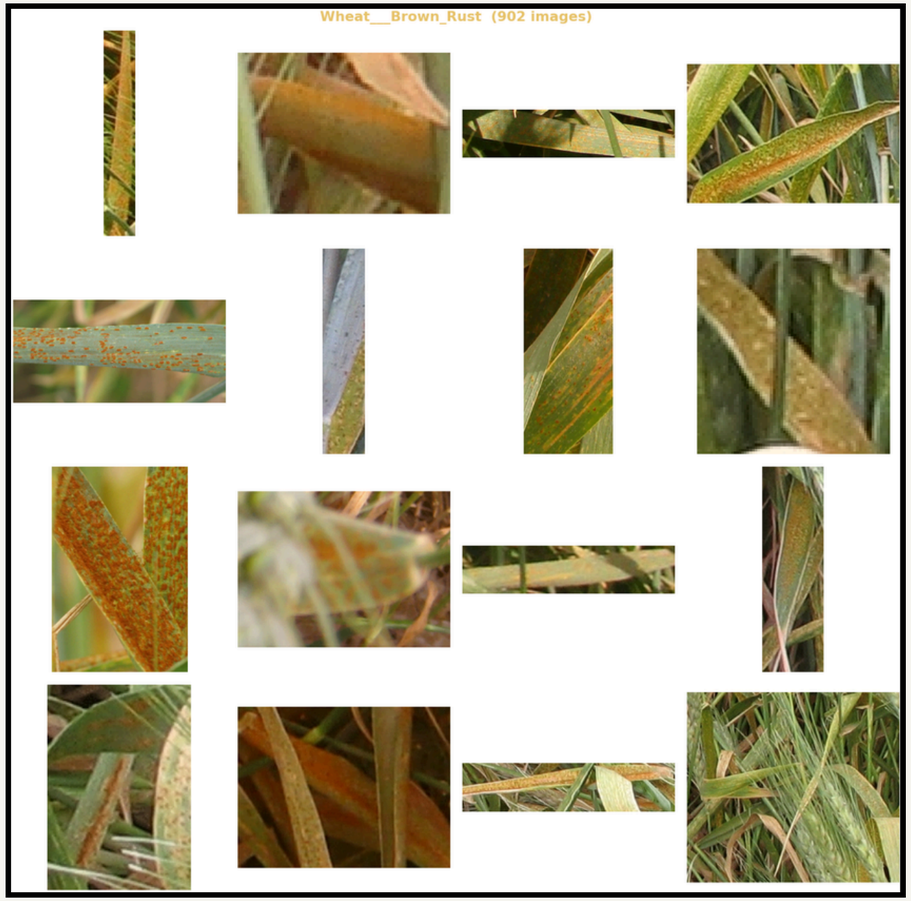
Crop	Classes
Corn	4 classes
Potato	3 classes
Rice	4 classes
Wheat	3 classes
Sugarcane	3 classes



Sugarcane severely under-represented — addressed via targeted augmentation

DATASET EXAMPLES





EXPLORATORY DATA ANALYSIS — KEY FINDINGS

Image Dimensions

Extreme variability:

16px → 6,000px width

Mandatory resize to 224×224

Color Channels

All 13,324 images confirmed

RGB (3-channel)

No grayscale or RGBA found

Color Analysis

Green channel dominant

in healthy leaves

Measurable shift in diseased classes

Duplicate Check

0 corrupt files

0 exact duplicates (MD5)

2 near-duplicate pairs (pHash)

Class Imbalance

Rice___Healthy: 1,488 imgs

Sugarcane classes: 100 imgs

Ratio: 14.9× imbalance

Data Quality

High overall quality

All files readable

Clean for model training

PREPROCESSING PIPELINE

1	2	3	4	5	6
Resize 224×224	**RGB Conversion**	**Duplicate Removal**	**Augmentation**	**Normalisation**	**Stratified Split**
LANCZOS resampling ImageNet standard input	Safety step retained 0 non-RGB found	MD5 exact + pHash near-duplicate removal	Minority classes only Flip · Rotate · Brightness	ImageNet mean/std For transfer learning	80 / 10 / 10 All 17 classes in each split

Augmentation applied ONLY to training images (never val/test) — prevents inflated evaluation metrics

SYSTEM ARCHITECTURE

End-to-end pipeline from leaf image to multilingual voice advisory

Leaf Image → **MobileNet Inference** → **Context Builder** → **ChromaDB Retriever** → **Groq LLM Generator** → **Voice Output**

Vision Module

MobileNet V3 Large (pretrained ImageNet)

Fine-tuned on 17 crop disease classes

92.1% test accuracy · 90% macro F1

Handles real-world image variability

RAG Pipeline

ICAR · TNAU · FAO agricultural docs

multilingual-e5-large embeddings

ChromaDB vector store (34+ chunks)

Metadata-filtered retrieval by crop+disease

Multimodal Output

Groq API — Llama 3.3 70B generation

6 Indian languages supported

Whisper ASR for voice input

gTTS for voice output

MODEL TRAINING — MobileNet V3 + Optuna HPO

Best Trial Parameters (Trial #4)	
Architecture	**MobileNet V3 Large**
Optimizer	**AdamW**
Learning Rate	**0.000428**
Dropout	**0.470**
Batch Size	**16**
Weight Decay	**1.8e-6**

92.1%
Test Accuracy

90.0%
Test Macro F1

0.207
Test Loss

92.5%
Val Macro F1

Optuna ran hyperparameter search across model architecture, optimizer, LR, dropout, batch size — MacroF1 as objective · MedianPruner for early stopping

RAG PIPELINE — KNOWLEDGE-GROUNDED ADVISORY

RAG = Retrieval Augmented Generation — LLM answers ONLY from retrieved verified documents, not from memory. Prevents hallucinated chemical names or wrong dosages reaching a farmer.

1. Ingest (One-time)

PDFs from ICAR · TNAU · FAO

PyMuPDF text extraction

Section-aware chunking

Filename → crop + disease metadata

multilingual-e5-large embeddings

Stored in ChromaDB (34 chunks)

2. Retrieve (Per Query)

Model outputs Corn___Gray_Leaf_Spot

Context Builder normalises to lowercase

Stage 1: Filter by crop + disease

Stage 2: Crop-only fallback

Stage 3: Unfiltered semantic search

Returns top 4 chunks as context

3. Generate (Groq LLM)

Llama 3.3 70B via Groq API (free)

System prompt: answer from context only

User language injected in prompt

Low confidence → KVK referral

No context → pre-translated fallback

gTTS converts response to audio

DEPLOYMENT

Hugging Face Spaces

Hosting (free)

Gradio UI · 16GB RAM · 2 vCPU

Public URL for demo

Groq API

LLM Inference (free)

Llama 3.3 70B · Fastest free tier

No credit card needed

ChromaDB

Vector Store

In-memory · Persisted to disk

34 agricultural doc chunks

Whisper

Speech-to-Text

whisper-small model

Indian language support

gTTS

Text-to-Speech

6 Indian languages

No API key needed

GitHub Actions

CI/CD

Auto-deploy to HF Spaces

on every push to main

What We Built

1. MobileNet V3 Large fine-tuned on 17 crop disease classes — 92.1% accuracy, 90% macro F1
2. RAG pipeline grounded in ICAR · TNAU · FAO documents — no hallucinated advice
3. Multilingual support in 6 Indian languages via voice input and output
4. Fully deployed on Hugging Face Spaces — zero cost, public demo URL
5. Modular architecture — each component independently testable and replaceable

Thank You